

Muhammad Irfan

Computer Science Undergraduate • AI & Systems Enthusiast • Fresher

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PROFESSIONAL SUMMARY

Motivated 2nd-year Computer Science student with a strong academic record (Dean's List) and hands-on experience across AI/Computer Vision, systems programming, and OOP. Hackathon winner with a proven ability to ship high-performance solutions under competition conditions. Seeking an entry-level software engineering or AI/ML role to apply research-grade skills in a real-world environment.

EDUCATION

Bachelor of Science in Computer Science

2024 – Present

FAST – NUCES, Karachi | CGPA: 3.73 / 4.0

- Dean's List: 2nd and 3rd Semesters — awarded for academic excellence
- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, OOP, Software Design & Analysis, Database Systems, Digital Logic Design, Discrete Mathematics, Computer Organization & Assembly Language

HONORS & ACHIEVEMENTS

- Runner Up — Computer Vision AI Hackathon (Duality x AIT): built a Secure Object Detection model achieving the highest mAP (Mean Average Precision) among all competing teams
- Image Segmentation Expert — successfully trained SegFormer MiT-B2 transformer model on Kaggle for semantic segmentation tasks
- Dean's List Certificate — awarded Semesters 2 and 3 for sustained academic excellence at FAST-NUCES

TECHNICAL SKILLS

Languages	C, C++, Python, SQL, x86 Assembly (COAL), HTML/CSS
AI & Vision	OpenCV, SegFormer (HuggingFace), YOLO-based Object Detection, Kaggle Notebooks
Frameworks	Qt Framework (C++), OOP Design Patterns
Tools	Git, GitHub, VS Code, Linux, Postman
Databases	SQL / PostgreSQL (basics), Firebase (basics)
Design & UML	UML (Class, Sequence, Activity, State, Component, Deployment), ECB, 4+1 View Model, Design Patterns
Soft Skills	Problem Solving, Hackathon Execution, Team Collaboration, Technical Documentation

PROJECTS

Flappy Bird Clone | C++, Qt Framework

2024

- Built a high-performance Flappy Bird clone using C++ and the Qt Framework with smooth animation and collision detection
- Implemented game loop, physics simulation, and score tracking using Qt's graphics and event systems
- Available on GitHub: github.com/MuhammadIrfannnnn/Flappy-Bird-Qt-CPP

Secure Object Detection Model (Hackathon Winner) | Python, YOLO, OpenCV

2025

- Developed an object detection pipeline that achieved the highest mAP score in the Duality x AIT Computer Vision Hackathon, placing 2nd overall
- Optimized inference speed and precision trade-off to maximize evaluation metrics under competition constraints
- Demonstrated low-level systems thinking with direct register and memory manipulation

Fast Navigation System | C++, Data Structures & Algorithms

2024

- Implemented an optimized pathfinding system using graph algorithms (Dijkstra/BFS) for fastest route computation

- Applied advanced DSA principles including priority queues and adjacency list representation

EXPERIENCE

Research & Development (Self-directed) | AI/CV Domain

2024 – Present

- Independently studied and applied state-of-the-art Computer Vision techniques including transformer-based segmentation and object detection
- Participated in competitive hackathon environment, delivering a production-quality model within a time-constrained setting
- Actively exploring LLM-OS integration — researching how large language models can interface with OS kernel architectures (in progress)

ONGOING / IN PROGRESS

- LLM-OS Architecture: researching integration of large language models with OS-level scheduling and process management
- Aviation Tracker: database-driven flight disruption analysis tool using SQL and real-world datasets

COURSES & SELF-LEARNING

- Kaggle: Computer Vision course — object detection, image segmentation, data augmentation
- HuggingFace Transformers: hands-on with SegFormer, BERT-family architectures
- GitHub Learning Lab: Git workflows, CI/CD fundamentals

References available upon request